



Top 53 CAT Averages, Mixtures, Alligations Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

A container has 40 liters of milk. Then, 4 liters are removed from the container and replaced with 4 liters of water. This process of replacing 4 liters of the liquid in the container with an equal volume of water is continued repeatedly. The smallest number of times of doing this process, after which the volume of milk in the container becomes less than that of water, is

A container has 40 liters of milk. Then, 4 liters are removed from the container and replaced with 4 liters of water. This process of replacing 4 liters of the liquid in the container with an equal volume of water is continued repeatedly. The smallest number of times of doing this process, after which the volume of milk in the container becomes less than that of water, is

Answer: _____

Handwritten calculations:

$$V \times \frac{V-z}{V}$$
$$1 \times \frac{40-4}{40} = 0.9$$
$$0.9 \times \frac{40-4}{40} = 0.81$$
$$0.81 \times \frac{40-4}{40} = 0.729$$
$$0.729 \times \frac{40-4}{40} = 0.6561$$
$$0.6561 \times \frac{40-4}{40} = 0.59049$$
$$0.59049 \times \frac{40-4}{40} = 0.531441$$
$$0.531441 \times \frac{40-4}{40} = 0.47829681$$

The video shows a man sitting at a desk with a laptop, explaining the solution.

VIDEO SOLUTION

Question 2

In a company, 20% of the employees work in the manufacturing department. If the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company, then the ratio of the average salary obtained by the manufacturing employees to the average salary obtained by the nonmanufacturing employees is

- A 6:5
- B 4:5
- C 5:4
- D 5:6

In a company, 20% of the employees work in the manufacturing department. If the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company, then the ratio of the average salary obtained by the manufacturing employees to the average salary obtained by the nonmanufacturing employees is

Options:

- A) 6:5
- B) 4:5
- C) 5:4
- D) 5:6

Handwritten calculations:

Total = 100
Manu = 20
Non Manu = 80

Ratio of average salary = $\frac{20}{80} \times \frac{100}{20} = \frac{1}{4} \times 5 = \frac{5}{4}$

The video shows a man sitting at a desk with a laptop, explaining the solution.

VIDEO SOLUTION

Question 3

There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by $2x$ kg. If the weight of E is 12 kg more than that of D, then the value of x is

- A 2
- B 0.5
- C 1
- D 1.5

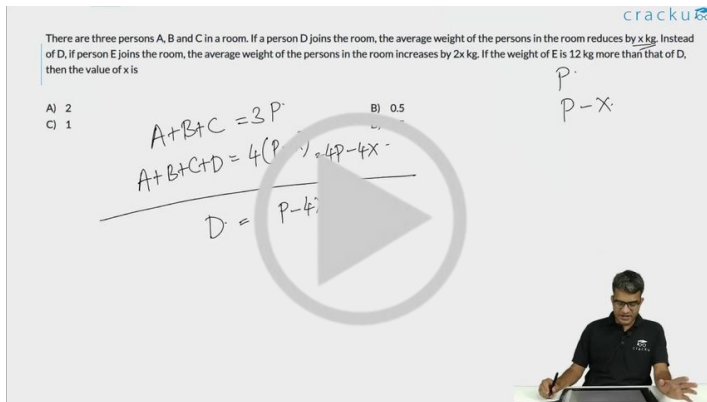
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There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by $2x$ kg. If the weight of E is 12 kg more than that of D, then the value of x is

A) 2
B) 0.5
C) 1

$$\begin{aligned} A+B+C &= 3P \\ A+B+C+D &= 4(P-x) = 4P-4x \\ \hline D &= P-4x \end{aligned}$$

P
 $P-x$



VIDEO SOLUTION

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Question 4

Anil mixes cocoa with sugar in the ratio 3 : 2 to prepare mixture A, and coffee with sugar in the ratio 7 : 3 to prepare mixture B. He combines mixtures A and B in the ratio 2 : 3 to make a new mixture C. If he mixes C with an equal amount of milk to make a drink, then the percentage of sugar in this drink will be

- A 17
- B 16
- C 21
- D 24

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Anil mixes cocoa with sugar in the ratio 3 : 2 to prepare mixture A, and coffee with sugar in the ratio 7 : 3 to prepare mixture B. He combines mixtures A and B in the ratio 2 : 3 to make a new mixture C. If he mixes C with an equal amount of milk to make a drink, then the percentage of sugar in this drink will be

A) 17
B) 16
C) 21

Handwritten notes: $\frac{3}{3+2} \times 10$, 10, 15, 2, 3, 7, 10, 5, 16.



VIDEO SOLUTION

Question 5

A mixture P is formed by removing a certain amount of coffee from a coffee jar and replacing the same amount with cocoa powder. The same amount is again removed from mixture P and replaced with same amount of cocoa powder to form a new mixture Q. If the ratio of coffee and cocoa in the mixture Q is 16 : 9, then the ratio of cocoa in mixture P to that in mixture Q is

- A 1 : 3
- B 1 : 2
- C 5 : 9
- D 4 : 9

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A mixture P is formed by removing a certain amount of coffee from a coffee jar and replacing the same amount with cocoa powder. The same amount is again removed from mixture P and replaced with same amount of cocoa powder to form a new mixture Q. If the ratio of coffee and cocoa in the mixture Q is 16 : 9, then the ratio of cocoa in mixture P to that in mixture Q is

A) 1 : 3
B) 1 : 2
C) 5 : 9

Handwritten notes: $(\frac{V-a}{V})^2$, $(\frac{V-a}{V})^3$, 0.9, 0.81, 0.729, 9, 16, 9, 16, 1:2.



VIDEO SOLUTION

Question 6

A glass contains 500 cc of milk and a cup contains 500 cc of water. From the glass, 150 cc of milk is transferred to the cup and mixed thoroughly. Next, 150 cc of this mixture is transferred from the cup to the glass. Now, the amount of water in the glass and the amount of milk in the cup are in the ratio

- A 1 : 1
- B 10 : 13
- C 3 : 10

D 10 : 3

A glass contains 500 cc of milk and a cup contains 500 cc of water. From the glass, 150 cc of milk is transferred to the cup and mixed thoroughly. Next, 150 cc of this mixture is transferred from the cup to the glass. Now, the amount of water in the glass and the amount of milk in the cup are in the ratio _____

A) 1 : 1
B) 10 : 13
C) 3 : 10
D) 10 : 3

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Question 7

There are two containers of the same volume, first container half-filled with sugar syrup and the second container half-filled with milk. Half the content of the first container is transferred to the second container, and then the half of this mixture is transferred back to the first container. Next, half the content of the first container is transferred back to the second container. Then the ratio of sugar syrup and milk in the second container is

- A 4 : 5
- B 6 : 5
- C 5 : 4
- D 5 : 6

There are two containers of the same volume, first container half-filled with sugar syrup and the second container half-filled with milk. Half the content of the first container is transferred to the second container, and then the half of this mixture is transferred back to the first container. Next, half the content of the first container is transferred back to the second container. Then the ratio of sugar syrup and milk in the second container is

A) 4 : 5
B) 6 : 5
C) 5 : 4
D) 5 : 6

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VIDEO SOLUTION

Question 8

A mixture contains lemon juice and sugar syrup in equal proportion. If a new mixture is created by adding this mixture and sugar syrup in the ratio 1 : 3, then the ratio of lemon juice and sugar syrup in the new mixture is

- A 1 : 4
- B 1 : 5
- C 1 : 6
- D 1 : 7

A mixture contains lemon juice and sugar syrup in equal proportion. If a new mixture is created by adding this mixture and sugar syrup in the ratio 1 : 3, then the ratio of lemon juice and sugar syrup in the new mixture is

A) 1:4
B) 1:5
C) 1:6
D) 1:7

(1) Original Mixture
(3) SS

VIDEO SOLUTION

Question 9

If a certain weight of an alloy of silver and copper is mixed with 3 kg of pure silver, the resulting alloy will have 90% silver by weight. If the same weight of the initial alloy is mixed with 2 kg of another alloy which has 90% silver by weight, the resulting alloy will have 84% silver by weight. Then, the weight of the initial alloy, in kg, is

- A 3.5
- B 2.5
- C 3
- D 4

If a certain weight of an alloy of silver and copper is mixed with 3 kg of pure silver, the resulting alloy will have 90% silver by weight. If the same weight of the initial alloy is mixed with 2 kg of another alloy which has 90% silver by weight, the resulting alloy will have 84% silver by weight. Then, the weight of the initial alloy, in kg, is

A) 3.5
B) 2.5
C) 3
D) 4

$3 + S = 90\%$
 $S = 90 - 3$

$2 + 1.8 = 84\%$

VIDEO SOLUTION

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Question 10

A person buys tea of three different qualities at ₹ 800, ₹ 500, and ₹ 300 per kg, respectively, and the amounts bought are in the proportion 2 : 3 : 5. She mixes all the tea and sells one-sixth of the mixture at ₹ 700 per kg. The price, in INR per kg, at which she should sell the remaining tea, to make an overall profit of 50%, is

- A 653
- B 688
- C 692
- D 675

cracku

A person buys tea of three different qualities at ₹ 800, ₹ 500, and ₹ 300 per kg, respectively, and the amounts bought are in the proportion 2 : 3 : 5. She mixes all the tea and sells one-sixth of the mixture at ₹ 700 per kg. The price, in INR per kg, at which she should sell the remaining tea, to make an overall profit of 50%, is

A) 653
C) 692

B) 688
D) 675

Handwritten calculations:
800 × 2 = 1600
500 × 3 = 1500
300 × 5 = 1500
Total = 4600
1/6 of mixture = 766.67 kg
766.67 × 700 = 5366.67
Remaining = 3833.33 kg
Total profit = 50%
Total revenue = 4600 × 1.5 = 6900
Price per kg = 6900 / 3833.33 = 688

VIDEO SOLUTION

Question 11

From a container filled with milk, 9 litres of milk are drawn and replaced with water. Next, from the same container, 9 litres are drawn and again replaced with water. If the volumes of milk and water in the container are now in the ratio of 16 : 9, then the capacity of the container, in litres, is

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From a container filled with milk, 9 litres of milk are drawn and replaced with water. Next, from the same container, 9 litres are drawn and again replaced with water. If the volumes of milk and water in the container are now in the ratio of 16 : 9, then the capacity of the container, in litres, is

Answer: _____

Handwritten calculations:
 $V \cdot \left(1 - \frac{9}{V}\right)^2 = \frac{16}{25} V$
 $V \cdot \left(\frac{V-9}{V}\right)^2 = \frac{16}{25} V$
 $\left(\frac{V-9}{V}\right)^2 = \frac{16}{25}$
 $\frac{V-9}{V} = \frac{4}{5}$
 $5(V-9) = 4V$
 $5V - 45 = 4V$
 $V = 45$

VIDEO SOLUTION

Question 12

Identical chocolate pieces are sold in boxes of two sizes, small and large. The large box is sold for twice the price of the small box. If the selling price per gram of chocolate in the large box is 12% less than that in the small box, then the percentage by which the weight of chocolate in the large box exceeds that in the small box is nearest to

- A 144

B 127

C 135

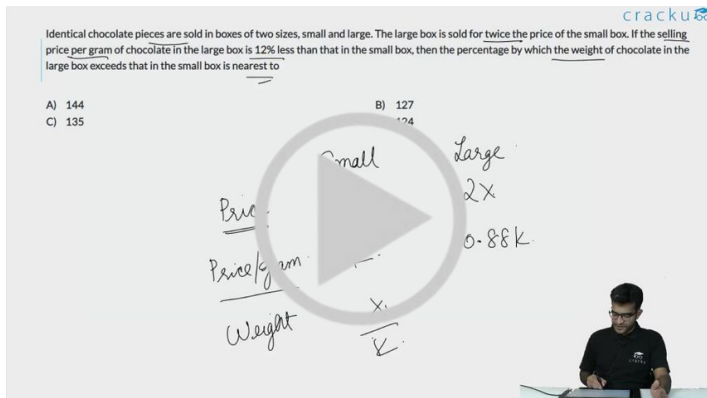
D 124

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Identical chocolate pieces are sold in boxes of two sizes, small and large. The large box is sold for twice the price of the small box. If the selling price per gram of chocolate in the large box is 12% less than that in the small box, then the percentage by which the weight of chocolate in the large box exceeds that in the small box is nearest to

A) 144
B) 127
C) 135

Small Large
Price $2x$
Price/gm $0.88x$
Weight x
 $\frac{x}{k}$



VIDEO SOLUTION



CAT Syllabus PDF



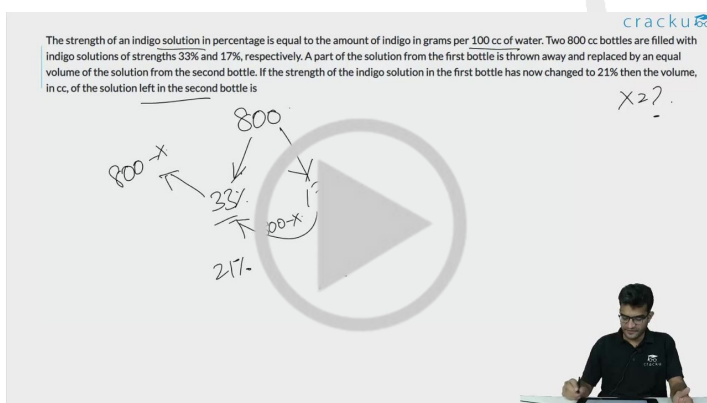
Question 13

The strength of an indigo solution in percentage is equal to the amount of indigo in grams per 100 cc of water. Two 800 cc bottles are filled with indigo solutions of strengths 33% and 17%, respectively. A part of the solution from the first bottle is thrown away and replaced by an equal volume of the solution from the second bottle. If the strength of the indigo solution in the first bottle has now changed to 21% then the volume, in cc, of the solution left in the second bottle is

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$800 - x$
 33%
 21%
 800
 17%
 x
 $x \approx ?$



VIDEO SOLUTION

Question 14

Two alcohol solutions, A and B, are mixed in the proportion 1:3 by volume. The volume of the mixture is then doubled by adding solution A such that the resulting mixture has 72% alcohol. If solution A has 60% alcohol, then the percentage of alcohol in solution B is

A 90%

- B 94%
- C 92%
- D 89%

Two alcohol solutions, A and B, are mixed in the proportion 1:3 by volume. The volume of the mixture is then doubled by adding solution A such that the resulting mixture has 72% alcohol. If solution A has 60% alcohol, then the percentage of alcohol in solution B is

VIDEO SOLUTION

Question 15

An alloy is prepared by mixing three metals A, B and C in the proportion 3 : 4 : 7 by volume. Weights of the same volume of the metals A, B and C are in the ratio 5 : 2 : 6. In 130 kg of the alloy, the weight, in kg, of the metal C is

- A 48
- B 84
- C 70
- D 96

An alloy is prepared by mixing three metals A, B and C in the proportion 3 : 4 : 7 by volume. Weights of the same volume of the metals A, B and C are in the ratio 5 : 2 : 6. In 130 kg of the alloy, the weight, in kg, of the metal C is

VIDEO SOLUTION

CAT Formula PDF

Question 16

A solution, of volume 40 litres, has dye and water in the proportion 2 : 3. Water is added to the solution to change this proportion to 2 : 5. If one fourths of this diluted solution is taken out, how many litres of dye must be added to the remaining solution to bring the proportion back to 2 : 3?

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A solution, of volume 40 litres, has dye and water in the proportion 2 : 3. Water is added to the solution to change this proportion to 2 : 5. If one fourths of this diluted solution is taken out, how many litres of dye must be added to the remaining solution to bring the proportion back to 2 : 3?

Handwritten notes:

- Initial ratio: $\frac{2}{3} = \frac{x}{40-x}$
- Initial dye: 16, Initial water: 24
- After adding water: $\frac{2}{5} = \frac{2}{24+x}$
- After removing 1/4: $\frac{2}{5} = \frac{2 \times \frac{3}{4}}{24 \times \frac{3}{4} + x}$
- Final dye: 16, Final water: 24



VIDEO SOLUTION

Question 17

A chemist mixes two liquids 1 and 2. One litre of liquid 1 weighs 1 kg and one litre of liquid 2 weighs 800 gm. If half litre of the mixture weighs 480 gm, then the percentage of liquid 1 in the mixture, in terms of volume, is

- A 80
- B 70
- C 85
- D 75

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A chemist mixes two liquids 1 and 2. One litre of liquid 1 weighs 1 kg, and one litre of liquid 2 weighs 800 gm. If half litre of the mixture weighs 480 gm, then the percentage of liquid 1 in the mixture, in terms of volume, is

Handwritten notes:

- 1 kg = 1000 gm, 800 gm
- 0.5 litre = 480 gm, 1 litre = 960 gm
- Let x be the volume of liquid 1, then (1-x) is the volume of liquid 2.
- Equation: $1000x + 800(1-x) = 960$
- Solving for x: $x = 0.8$
- Percentage of liquid 1: 80%



VIDEO SOLUTION

Question 18

The strength of a salt solution is p% if 100 ml of the solution contains p grams of salt. Each of three vessels A, B, C contains 500 ml of salt solution of strengths 10%, 22%, and 32%, respectively. Now, 100 ml of the solution in vessel A is transferred to vessel B. Then, 100 ml of the solution in vessel B is transferred to vessel C. Finally, 100 ml of the solution in vessel C is transferred to vessel A. The strength, in percentage, of the resulting solution in vessel A is

- A 15
- B 13

C 12

D 14

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The strength of a salt solution is $p\%$ if 100 ml of the solution contains p grams of salt. Each of three vessels A, B, C contains 500 ml of salt solution of strengths 10%, 22%, and 32%, respectively. Now, 100 ml of the solution in vessel A is transferred to vessel B. Then, 100 ml of the solution in vessel B is transferred to vessel C. Finally, 100 ml of the solution in vessel C is transferred to vessel A. The strength, in percentage, of the resulting solution in vessel A is

A) 15
B) 12
C) 12
D) 14

A: 500 ml, 10%
B: 500 ml, 22%
C: 500 ml, 32%

Transfer 100 ml from A to B:
A: 400 ml, 10%
B: 600 ml, 16.67%

Transfer 100 ml from B to C:
B: 500 ml, 22.22%
C: 600 ml, 28.89%

Transfer 100 ml from C to A:
A: 500 ml, 12%

VIDEO SOLUTION

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Question 19

The average weight of students in a class increases by 600 gm when some new students join the class. If the average weight of the new students is 3 kg more than the average weight of the original students, then the ratio of the number of original students to the number of new students is

A 1 : 4

B 1 : 2

C 4 : 1

D 3 : 1

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The average weight of students in a class increases by 600 gm when some new students join the class. If the average weight of the new students is 3 kg more than the average weight of the original students, then the ratio of the number of original students to the number of new students is

A) 1 : 4
B) 1 : 2
C) 4 : 1
D) 3 : 1

A: w
B: $w+2$
A+B: $w+0.6$

VIDEO SOLUTION

Question 20

Manu earns ₹4000 per month and wants to save an average of ₹550 per month in a year. In the first nine months, his monthly expense was ₹3500, and he foresees that, tenth month onward, his monthly expense will increase to ₹3700. In order to meet his yearly savings target, his monthly earnings, in rupees, from the tenth

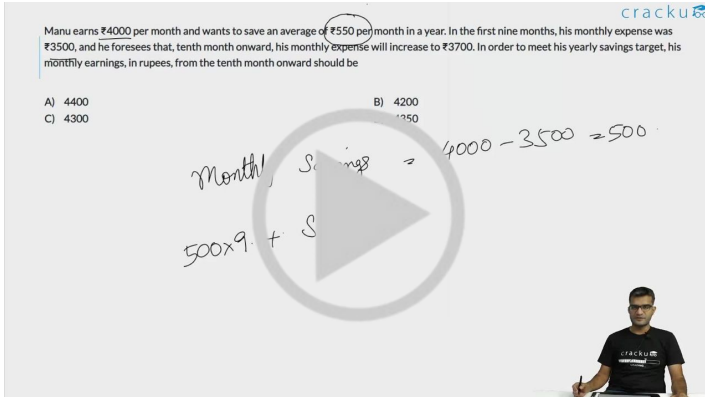
month onward should be

- A 4400
- B 4200
- C 4300
- D 4350

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A) 4400
B) 4200
C) 4300
D) 4350

Month! Savings $\rightarrow 4000 - 3500 = 500$
 $500 \times 9 + S$



VIDEO SOLUTION

Question 21

In an examination, the average marks of students in sections A and B are 32 and 60, respectively. The number of students in section A is 10 less than that in section B. If the average marks of all the students across both the sections combined is an integer, then the difference between the maximum and minimum possible number of students in section A is

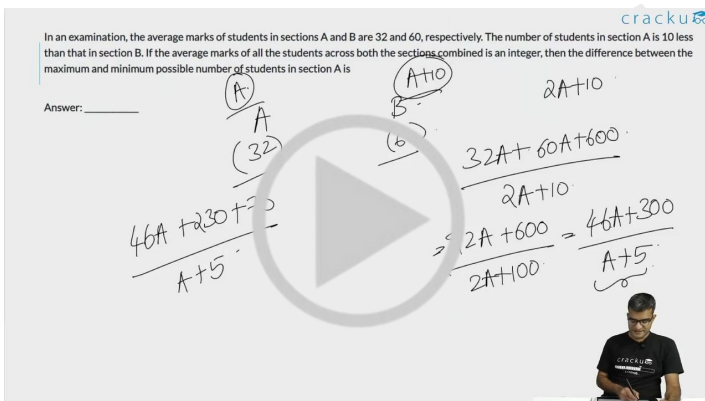
In an examination, the average marks of students in sections A and B are 32 and 60, respectively. The number of students in section A is 10 less than that in section B. If the average marks of all the students across both the sections combined is an integer, then the difference between the maximum and minimum possible number of students in section A is

Answer: _____

A (A)
A (32)
 $\frac{46A + 230 + 70}{A + 5}$

B $(A+10)$
B (6)
 $\frac{32A + 60A + 600}{2A + 10}$

$\frac{46A + 230 + 70}{A + 5} = \frac{32A + 60A + 600}{2A + 10}$
 $\frac{46A + 230 + 70}{A + 5} = \frac{46A + 300}{A + 5}$



VIDEO SOLUTION

CAT Percentile Predictor



Question 22

The average of three integers is 13. When a natural number n is included, the average of these four integers remains an odd integer. The minimum possible value of n is

- A 3

- B 4
- C 5
- D 1

The average of three integers is 13. When a natural number n is included, the average of these four integers remains an odd integer. The minimum possible value of n is

A) 3
C) 5

B) 4
D) 1

$13 \times 3 = 39$

$\frac{39+n}{4} = \text{odd integer}$

VIDEO SOLUTION

Question 23

If a and b are non-negative real numbers such that $a + 2b = 6$, then the average of the maximum and minimum possible values of $(a + b)$ is

- A 3
- B 4
- C 3.5
- D 4.5

If a and b are non-negative real numbers such that $a + 2b = 6$, then the average of the maximum and minimum possible values of $(a + b)$ is

A) 3
C) 3.5

B) 4
D) 4.5

$a + b = 6 - b$

$(a+b)_{\max} = 6$
 $b_{\min} = 0$

VIDEO SOLUTION

Question 24

A batsman played $n + 2$ innings and got out on all occasions. His average score in these $n + 2$ innings was 29 runs and he scored 38 and 15 runs in the last two innings. The batsman scored less than 38 runs in each of the first n innings. In these n innings, his average score was 30 runs and lowest score was x runs. The smallest possible value of x is

- A 4

B 3

C 2

D 1

A batsman played $n+2$ innings and got out on all occasions. His average score in these $n+2$ innings was 29 runs and he scored 38 and 15 runs in the last two innings. The batsman scored less than 38 runs in each of the first n innings. In these n innings, his average score was 30 runs and lowest score was x runs. The smallest possible value of x is

Handwritten solution:

$$\text{Total score} = (n+2) \times 29$$
$$= 50 + 38 + 15$$
$$29 + 58 = n + 53$$
$$n = 15$$

Diagram: A large play button icon is overlaid on the solution. Below it, a small video player shows a person writing.

VIDEO SOLUTION

 **CAT Score Calculator**

Question 25

In a football tournament, a player has played a certain number of matches and 10 more matches are to be played. If he scores a total of one goal over the next 10 matches, his overall average will be 0.15 goals per match. On the other hand, if he scores a total of two goals over the next 10 matches, his overall average will be 0.2 goals per match. The number of matches he has played is

Handwritten solution:

$$\frac{G+1}{N+10} = 0.15$$
$$\frac{G+2}{N+10} = 0.2$$

Diagram: A large play button icon is overlaid on the solution. Below it, a small video player shows a person writing.

VIDEO SOLUTION

Question 26

The arithmetic mean of scores of 25 students in an examination is 50. Five of these students top the examination with the same score. If the scores of the other students are distinct integers with the lowest being 30, then the maximum possible score of the toppers is

The arithmetic mean of scores of 25 students in an examination is 50. Five of these students top the examination with the same score. If the scores of the other students are distinct integers with the lowest being 30, then the maximum possible score of the toppers is

Answer: _____

Handwritten solution:

Let the score of the toppers be x . Then the scores of the other 20 students are 30, 31, 32, ..., 49. The sum of these scores is $20 \times 34.5 = 690$. The total sum of scores is $25 \times 50 = 1250$. Therefore, $5x + 690 = 1250$, which gives $5x = 560$ and $x = 112$.

Video solution button: VIDEO SOLUTION

Question 27

Onion is sold for 5 consecutive months at the rate of Rs 10, 20, 25, 25, and 50 per kg, respectively. A family spends a fixed amount of money on onion for each of the first three months, and then spends half that amount on onion for each of the next two months. The average expense for onion, in rupees per kg, for the family over these 5 months is closest to

- A 26
- B 18
- C 16
- D 20

Onion is sold for 5 consecutive months at the rate of Rs 10, 20, 25, 25, and 50 per kg, respectively. A family spends a fixed amount of money on onion for each of the first three months, and then spends half that amount on onion for each of the next two months. The average expense for onion, in rupees per kg, for the family over these 5 months is closest to

Options: A) 26, B) 18, C) 16

Handwritten solution:

Let the fixed amount spent in the first three months be $100F$. Then the amount spent in the next two months is $50F$. The total amount spent is $150F$. The total weight of onion bought is $10 + 20 + 25 + 25 + 50 = 130$ kg. The average expense per kg is $\frac{150F}{130} \approx 11.54$.

Video solution button: VIDEO SOLUTION

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Question 28

Five students, including Amit, appear for an examination in which possible marks are integers between 0 and 50, both inclusive. The average marks for all the students is 38 and exactly three students got more than 32. If no two students got the same marks and Amit got the least marks among the five students, then the difference between the highest and lowest possible marks of Amit is

- A 22
- B 21
- C 24
- D 20

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Five students, including Amit, appear for an examination in which possible marks are integers between 0 and 50, both inclusive. The average marks for all the students is 38 and exactly three students got more than 32. If no two students got the same marks and Amit got the least marks among the five students, then the difference between the highest and lowest possible marks of Amit is

A) 22
B) 21
C) 24
D) 20

min for Amit

48
32 17 30 20
50 + 49 + 32 = 131
80

Total Marks = $38 \times 5 = 190$

2170

VIDEO SOLUTION

Question 29

The average of 30 integers is 5. Among these 30 integers, there are exactly 20 which do not exceed 5. What is the highest possible value of the average of these 20 integers?

- A 3.5
- B 5
- C 4.5
- D 4

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The average of 30 integers is 5. Among these 30 integers, there are exactly 20 which do not exceed 5. What is the highest possible value of the average of these 20 integers?

Average of 30 integers = 5
Total = $30 \times 5 = 150$

$20 \leq 5$
 $10 > 5$
 $10 \geq 6$

VIDEO SOLUTION

Question 30

The average marks of a student in 10 papers are 80. If the highest and the lowest scores are not considered, the average is 81. If his highest score is 92, find the lowest.

- A 55

- B 60
- C 62
- D Cannot be determined

The average marks of a student in 10 papers are 80. If the highest and the lowest scores are not considered, the average is 81. If his highest score is 92, find the lowest.

A) 55
B) 60
C) 62
D) Cannot be determined

Handwritten solution:

$$\text{Total} = 80 \times 10 = 800$$

$$H, L, \text{ and } 8 \times 81 = 648$$

$$800 - 648 = 152$$

$$92 - 152 = -60$$

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Question 31

Total expenses of a boarding house are partly fixed and partly varying linearly with the number of boarders. The average expense per boarder is Rs. 700 when there are 25 boarders and Rs. 600 when there are 50 boarders. What is the average expense per boarder when there are 100 boarders?

- A 550
- B 580
- C 540
- D 560

Total expenses of a boarding house are partly fixed and partly varying linearly with the number of boarders. The average expense per boarder is Rs. 700 when there are 25 boarders and Rs. 600 when there are 50 boarders. What is the average expense per boarder when there are 100 boarders?

A) 550
B) 580
C) 540
D) 560

Handwritten solution:

$$TC = F + nV$$

$$17500 = F + 25V$$

$$30000 = F + 50V$$

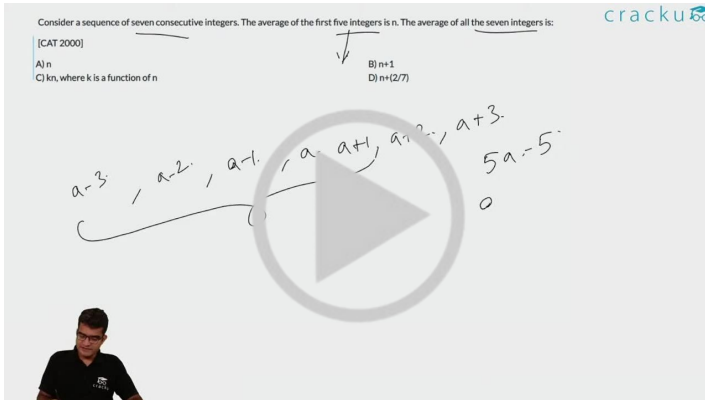
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VIDEO SOLUTION

Question 32

Consider a sequence of seven consecutive integers. The average of the first five integers is n . The average of all the seven integers is:

- A n
- B $n+1$
- C kn , where k is a function of n
- D $n+(2/7)$



 VIDEO SOLUTION

Question 33

Three classes X, Y and Z take an algebra test.

The average score in class X is 83.

The average score in class Y is 76.

The average score in class Z is 85.

The average score of all students in classes X and Y together is 79.

The average score of all students in classes Y and Z together is 81.

What is the average for all the three classes?

- A 81
- B 81.5
- C 82
- D 84.5

Three classes X, Y and Z take an algebra test.
 The average score in class X is 83.
 The average score in class Y is 76.
 The average score in class Z is 85.
 The average score of all students in classes X and Y together is 79.
 The average score of all students in classes Y and Z together is 81.
 What is the average for all the three classes?

A) 81
 B) 82

Handwritten solution for Question 34: A tree diagram shows the calculation of the average for three classes. Class X has an average of 83, Class Y has 76, and Class Z has 85. The average of X and Y is 79, and the average of Y and Z is 81. The final average for all three classes is calculated as 81.5, which is rounded to 82.

VIDEO SOLUTION

CAT Daily Targets
 Step by step, heading towards 99+ percentile

Question 34

Amol was asked to calculate the arithmetic mean of 10 positive integers, each of which had 2 digits. By mistake, he interchanged the 2 digits, say a and b , in one of these 10 integers. As a result, his answer for the arithmetic mean was 1.8 more than what it should have been. Then $|b - a|$ equals

- A 1
- B 2
- C 3
- D None of these

Amol was asked to calculate the arithmetic mean of 10 positive integers, each of which had 2 digits. By mistake, he interchanged the 2 digits, say a and b , in one of these 10 integers. As a result, his answer for the arithmetic mean was 1.8 more than what it should have been. Then $|b - a|$ equals

A) 1
 B) 2
 C) 3
 D) None of these

Handwritten solution for Question 35: Let the original number be $10a + b$ and the interchanged number be $10b + a$. The difference between the two numbers is $9b - 9a = 9(b - a)$. This difference is equal to 18 (since the mean increased by 1.8 for 10 numbers). So, $9(b - a) = 18$, which gives $b - a = 2$.

VIDEO SOLUTION

Question 35

Ten years ago, the ages of the members of a joint family of eight people added up to 231 years. Three years later, one member died at the age of 60 years and a child was born during the same year. After another three years, one more member died, again at 60, and a child was born during the same year. The current average age of this eight-member joint family is nearest to

- A 23 years

- B 22 years
- C 21 years
- D 25 years
- E 24 years

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[CAT 2007]

A) 23 years
C) 21 years
E) 24 years

B) 22 years
D) 25 years

VIDEO SOLUTION

Question 36

Consider the set $S = \{2, 3, 4, \dots, 2n+1\}$, where n is a positive integer larger than 2007. Define X as the average of the odd integers in S and Y as the average of the even integers in S . What is the value of $X - Y$?

- A 0
- B 1
- C $(1/2)*n$
- D $(n+1)/2n$
- E 2008

Consider the set $S = \{2, 3, 4, \dots, 2n+1\}$, where n is a positive integer larger than 2007. Define X as the average of the odd integers in S and Y as the average of the even integers in S . What is the value of $X - Y$?

A) 0
C) $(1/2)*n$
E) 2008

B) 1
D) $(n+1)/2n$

VIDEO SOLUTION

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Question 37

A class consists of 20 boys and 30 girls. In the mid-semester examination, the average score of the girls was 5 higher than that of the boys. In the final exam, however, the average score of the girls dropped by 3 while the average score of the entire class increased by 2. The increase in the average score of the boys is

- A 9.5
- B 10
- C 4.5
- D 6

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A class consists of 20 boys and 30 girls. In the mid-semester examination, the average score of the girls was 5 higher than that of the boys. In the final exam, however, the average score of the girls dropped by 3 while the average score of the entire class increased by 2. The increase in the average score of the boys is

Handwritten solution:

Let the average score of boys in the mid-semester exam be A .
Then the average score of girls is $A + 5$.
Total score in mid = $20A + 30(A + 5) = 20A + 30A + 150 = 50A + 150$
Average score in mid = $\frac{50A + 150}{50} = A + 3$

In the final exam, the average score of girls dropped by 3, so it is $A + 5 - 3 = A + 2$.
Let the increase in the average score of boys be x .
Then the average score of boys in the final exam is $A + x$.
Total score in final = $20(A + x) + 30(A + 2) = 20A + 20x + 30A + 60 = 50A + 20x + 60$
Average score in final = $\frac{50A + 20x + 60}{50} = A + \frac{20x}{50} + \frac{60}{50} = A + \frac{2x}{5} + 1.2$

Since the average score of the entire class increased by 2, we have:
 $A + 3 + 2 = A + \frac{2x}{5} + 1.2$
 $5 = \frac{2x}{5} - 0.2$
 $5.2 = \frac{2x}{5}$
 $x = \frac{5.2 \times 5}{2} = 13$

Wait, the correct answer is 9.5. Let's re-evaluate the final average score calculation.

Correct calculation for final average score:
Total score in final = $20(A + x) + 30(A + 2) = 20A + 20x + 30A + 60 = 50A + 20x + 60$
Average score in final = $\frac{50A + 20x + 60}{50} = A + \frac{2x}{5} + 1.2$

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 $x = \frac{5.2 \times 5}{2} = 13$

VIDEO SOLUTION

Question 38

The average height of 22 toddlers increases by 2 inches when two of them leave this group. If the average height of these two toddlers is one-third the average height of the original 22, then the average height, in inches, of the remaining 20 toddlers is

- A 30
- B 28
- C 32
- D 26

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The average height of 22 toddlers increases by 2 inches when two of them leave this group. If the average height of these two toddlers is one-third the average height of the original 22, then the average height, in inches, of the remaining 20 toddlers is

Handwritten solution:

Let the average height of the original 22 toddlers be A .
Total height of 22 toddlers = $22A$

When two toddlers leave, the average height of the remaining 20 toddlers increases by 2 inches, so it is $A + 2$.
Total height of 20 toddlers = $20(A + 2) = 20A + 40$

The average height of the two toddlers who left is $\frac{A}{3}$.
Total height of the two toddlers = $2 \times \frac{A}{3} = \frac{2A}{3}$

Since the total height of the 20 toddlers plus the height of the two toddlers who left equals the total height of the original 22 toddlers, we have:
 $20A + 40 + \frac{2A}{3} = 22A$
 $40 + \frac{2A}{3} = 2A$
 $40 = 2A - \frac{2A}{3} = \frac{4A}{3}$
 $A = \frac{40 \times 3}{4} = 30$

VIDEO SOLUTION

Question 39

An elevator has a weight limit of 630 kg. It is carrying a group of people of whom the heaviest weighs 57 kg and the lightest weighs 53 kg. What is the maximum possible number of people in the group?

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An elevator has a weight limit of 630 kg. It is carrying a group of people of whom the heaviest weighs 57 kg and the lightest weighs 53 kg. What is the maximum possible number of people in the group?

Heaviest = 57 kg
Lightest = 53 kg

$$n \leq 630 - 110 = 520$$



 VIDEO SOLUTION

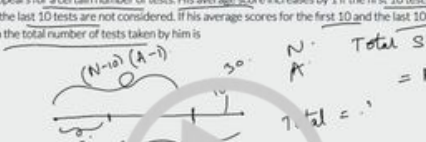
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Question 40

A CAT aspirant appears for a certain number of tests. His average score increases by 1 if the first 10 tests are not considered, and decreases by 1 if the last 10 tests are not considered. If his average scores for the first 10 and the last 10 tests are 20 and 30, respectively, then the total number of tests taken by him is

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The diagram shows a number line with points marked at 10, 20, and 30. Above the line, there are labels $(N-10)$ and $(A-1)$ near the 10 mark, and $(N-1)$ near the 30 mark. To the right of the diagram, the formula for total score is written as:

$$\frac{N \cdot A}{N} = \frac{\text{Total Score}}{N}$$

Below the formula, it says "Total =".



 VIDEO SOLUTION

Question 41

In an apartment complex, the number of people aged 51 years and above is 30 and there are at most 39 people whose ages are below 51 years. The average age of all the people in the apartment complex is 38 years. What is the largest possible average age, in years, of the people whose ages are below 51 years?

- | | |
|----------|----|
| A | 27 |
| B | 25 |
| C | 26 |

D 28

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In an apartment complex, the number of people aged 51 years and above is 30 and there are at most 39 people whose ages are below 51 years. The average age of all the people in the apartment complex is 38 years. What is the largest possible average age, in years, of the people whose ages are below 51 years?

$$P \geq 51 \Rightarrow 30$$

$$Q < 51 \Rightarrow 39 \cdot x$$

$$\frac{30 \cdot 51 + 39 \cdot x}{30 + 39} = 38$$

$$P \geq 51$$

$$Q < 51$$

$$x \leq 39$$

Total

38



VIDEO SOLUTION

Question 42

Ramesh and Gautam are among 22 students who write an examination. Ramesh scores 82.5. The average score of the 21 students other than Gautam is 62. The average score of all the 22 students is one more than the average score of the 21 students other than Ramesh. The score of Gautam is

- A 53
- B 51
- C 48
- D 49

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Ramesh and Gautam are among 22 students who write an examination. Ramesh scores 82.5. The average score of the 21 students other than Gautam is 62. The average score of all the 22 students is one more than the average score of the 21 students other than Ramesh. The score of Gautam is

$$82.5$$

$$62$$

$$22 \times (a+1) = 21 \times a$$



VIDEO SOLUTION

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CAT Daily Articles




Question 43

There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is

There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is

Answer: _____

Handwritten notes on the screen:

$$\begin{aligned} \frac{A+B}{2} &= A+1 \\ A+B &= 2A+2 \\ B &= A+2 \end{aligned}$$

Diagram showing four numbers A, B, C, D in increasing order. Brackets indicate: $A+B = 2(A+1)$, $A+B+C = 2(A+B) + 2$, and $A+B+C+D = 2(A+B+C) + 2$.

VIDEO SOLUTION

Question 44

The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64. Then, the largest number in the original set of three numbers is

The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64. Then, the largest number in the original set of three numbers is

Answer: _____

Handwritten notes on the screen:

$$\begin{aligned} a < b < c \\ a+7 &< b < c-10 \\ a+7+b+c-10 &= b' \\ \frac{a+b+c-3}{3} &= 3(b) \end{aligned}$$

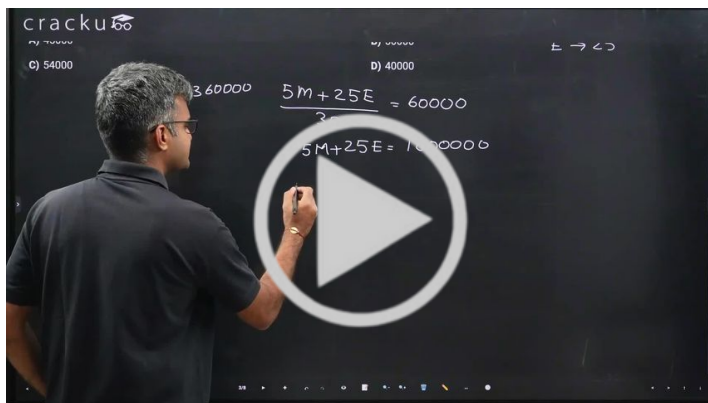
Equation derived: $a+b+c = 84 \quad \text{--- (1)}$

VIDEO SOLUTION

Question 45

The average salary of 5 managers and 25 engineers in a company is 60000 rupees. If each of the managers received 20% salary increase while the salary of the engineers remained unchanged, the average salary of all 30 employees would have increased by 5%. The average salary, in rupees, of the engineers is

- A 45000
- B 50000
- C 54000
- D 40000



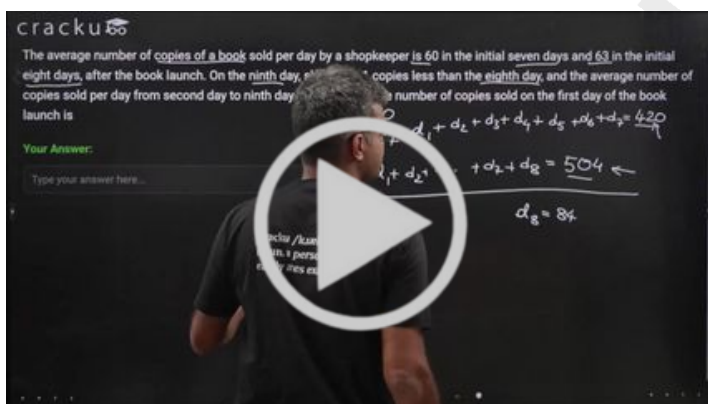
VIDEO SOLUTION

100+ CAT Quant Questions



Question 46

The average number of copies of a book sold per day by a shopkeeper is 60 in the initial seven days and 63 in the initial eight days, after the book launch. On the ninth day, she sells 11 copies less than the eighth day, and the average number of copies sold per day from second day to ninth day becomes 66. The number of copies sold on the first day of the book launch is



VIDEO SOLUTION

Question 47

A glass is filled with milk. Two-thirds of its content is poured out and replaced with water. If this process of pouring out two-thirds the content and replacing with water is repeated three more times, then the final ratio of milk to water in the glass, is

- A 1 : 27
- B 1 : 80
- C 1 : 81
- D 1 : 26



B 7 : 9

C 3 : 7

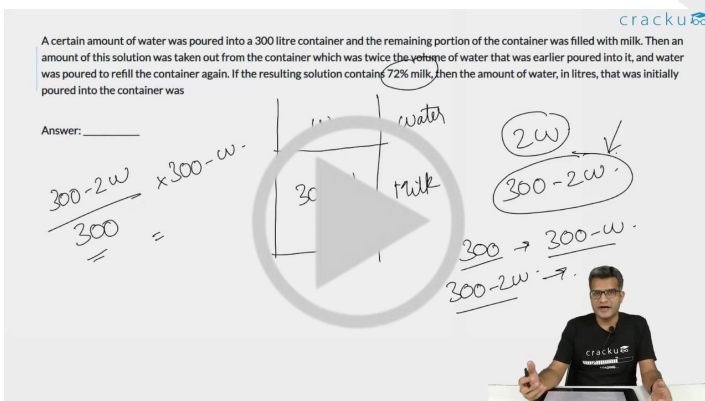
D 1 : 1



VIDEO SOLUTION

Question 50

A certain amount of water was poured into a 300 litre container and the remaining portion of the container was filled with milk. Then an amount of this solution was taken out from the container which was twice the volume of water that was earlier poured into it, and water was poured to refill the container again. If the resulting solution contains 72% milk, then the amount of water, in litres, that was initially poured into the container was



VIDEO SOLUTION

Question 51

Vessels A and B contain 60 litres of alcohol and 60 litres of water, respectively. A certain volume is taken out from A and poured into B. After stirring, the same volume is taken out from B and poured into A. If the resultant ratio of alcohol and water in A is 15 : 4, then the volume, in litres, initially taken out from A is



VIDEO SOLUTION

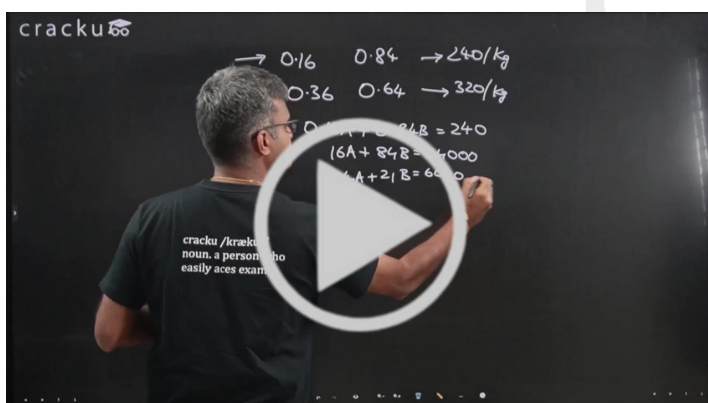
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Question 52

A mixture of coffee and cocoa, 16% of which is coffee, costs Rs 240 per kg. Another mixture of coffee and cocoa, of which 36% is coffee, costs Rs 320 per kg. If a new mixture of coffee and cocoa costs Rs 376 per kg, then the quantity, in kg, of coffee in 10 kg of this new mixture is

- A 5
- B 4
- C 2.5
- D 6



VIDEO SOLUTION

Question 53

A container holds 200 litres of a solution of acid and water, having 30% acid by volume. Atul replaces 20% of this solution with water, then replaces 10% of the resulting solution with acid, and finally replaces 15% of the solution thus obtained, with water. The percentage of acid by volume in the final solution obtained after these three replacements, is nearest to

- A 23

B 25

C 29

D 27

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water. The percentage of acid by volume in the final solution obtained after these three replacements, is nearest to

A) 23 B) 25 C) 29 D) 27

Mixture
Acid *Water*
 $A\%$ $(100-A)\%$
 10% $(100-A) \times 0.9$
 20% $(100-A) \times 0.8$
acid



 VIDEO SOLUTION



Answers

1.7	2.B	3.C	4.A	5.C	6.A	7.D	8.D
9.C	10.B	11.45	12.B	13.200	14.C	15.B	16.8
17.A	18.D	19.C	20.A	21.63	22.C	23.D	24.C
25.10	26.92	27.B	28.D	29.C	30.B	31.A	32.B
33.B	34.B	35.E	36.B	37.A	38.C	39.11	40.60
41.D	42.B	43.15	44.70	45.C	46.49	47.B	48.B
49.B	50.30	51.16	52.A	53.D			

Explanations

1.7

Let's assume that after n iteration, the volume of the milk will be less than 50%, which is less than 20 liters.

Initially, the amount of milk is 40 liters, after the first iteration, the volume of milk is $40 \cdot \frac{9}{10}$

After the second iteration, the volume of milk is $40 \times \left(\frac{9}{10}\right)^2$

Similarly, after the n iterations, the volume of milk is $40 \times \left(\frac{9}{10}\right)^n$

Now,

$$40 \times \left(\frac{9}{10}\right)^n \leq 20$$

$$\Rightarrow \left(\frac{9}{10}\right)^n \leq \frac{1}{2}$$

$$\Rightarrow n \geq 7$$

Hence, the correct answer is 7

[VIDEO SOLUTION](#)

2.B

Let the number of total employees in the company be $100x$, and the total salary of all the employees be $100y$.

It is given that 20% of the employees work in the manufacturing department, and the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company.

Hence, the total number of employees in the manufacturing department is $20x$, and the total salary received by them is $(100y/6)$

Average salary in the manufacturing department = $(100y/6 \cdot 20x) = 5y/6x$

Similarly, the total number of employees in the nonmanufacturing department is $80x$, and the total salary received by them is $(500y/6)$

Hence, the average salary in the nonmanufacturing department = $(500y/6 \cdot 80x) = 25y/24x$

Hence, the ratio is:- $(5y/6x) : (25y/24x)$

$$\Rightarrow 120 : 150 = 4 : 5$$

The correct option is B

[VIDEO SOLUTION](#)

3.C

Let us assume that A, B, C, D, and E weights are a , b , c , d , and e .

1st condition

$$\frac{(a+b+c)}{3} - \frac{(a+b+c+d)}{4} = x$$

2nd condition

$$\frac{(a+b+c+e)}{4} - \frac{(a+b+c)}{3} = 2x$$

Adding both the equations, we get:

$$\frac{(e-d)}{4} = 3x$$

$$\Rightarrow \frac{(e-d)}{4} = 3x \Rightarrow e - d = 12x$$

Given that $12x = 12 \Rightarrow x = 1$.

 VIDEO SOLUTION

4. A

Let the volume of mixture A be 200 ml, which implies the quantity of cocoa in the mixture is 120 ml, and the quantity of sugar in the mixture 80 ml.

Similarly, let the volume of the mixture be 300 ml, which implies the quantity of coffee, and sugar in the mixture is 210, and 90 ml, respectively.

Now we combine mixture A, and B in the ratio of 2:3 (if 200 ml mixture A, then 300 ml of mixture B).

Hence, the volume of the mixture C is $(200+300) = 500$ ml, and the quantity of the sugar is $(90+80) = 170$ ml.

Now he mixes C with an equal amount of milk to make a drink, which implies the quantity of the final mixture is $(500+500) = 1000$ ml.

The quantity of sugar in the final mixture is 170 ml.

Hence, the percentage is 17%

The correct option is A

 VIDEO SOLUTION

5. C

Given that in the final mixture, the ratio of coffee and cocoa is 16:9

Let us assume coffee is 16 units and cocoa is 9 units.

$\Rightarrow$ Initially, there are 25 units of coffee and 0 units of cocoa

Let's say x units of the mixture is removed and replaced with cocoa

$\Rightarrow$ Now, we have $(25-x)$ coffee and x units of cocoa. $\Rightarrow$ Mixture P

Now, if x units of the mixture is removed:

$$\text{Amount of coffee present} = (25-x) - \frac{(25-x)}{25} \times x$$

$$\Rightarrow (25-x) \left(1 - \frac{x}{25}\right) = 16$$

$$\Rightarrow (25-x)^2 = 16 \times 25$$

$$\Rightarrow 25 - x = 20 \Rightarrow x = 5.$$

In mixture P, cocoa = $x = 5$

In mixture Q, cocoa = 9 units.

$\Rightarrow$ Required ratio = 5:9

 VIDEO SOLUTION

CAT Expected Questions PDF

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6.A

Initially: a glass 500cc milk and a cup 500cc water

Step 1: 150 cc of milk is transferred to the cup from glass

After step 1: Glass - 350 cc milk, Cup - 150 cc milk and 500 cc water

Step 2: 150 cc of this mixture is transferred from the cup to the glass

After step 2:

Glass - 350 cc milk + 150 cc mixture with milk:water ratio 3:10

Cup - 500 cc mixture with milk:water ratio 3:10

$$\text{water in glass : milk in cup} = \frac{10}{13} \times 150 : \frac{3}{13} \times 500 = 1 : 1$$

The answer is option A.

VIDEO SOLUTION

7.D

Initially	1st-Sugar syrup-100L	2nd - Milk - 100L
After step 1	Sugar Syrup - 50L	SS- 50L, Milk - 100L
After step 2	SS - 50+25 = 75L Milk - 50L	SS - 25L, Milk - 50L
After step 3	Milk - 25L SS - 37.5L	Milk - 50 + 25 = 75L SS - 25 + 37.5 = 62.5L

Step 1: Half the content of the first container is transferred to the second container

Step 2: Half of the mixture of second container is transferred back to the first container

Step 3: Half the content of the first container is transferred back to the second container

Sugar syrup : Milk in second container = 62.5 : 75 = 5 : 6

The answer is option D.

VIDEO SOLUTION

8.D

Lemon juice : sugar syrup in the mixture is 1:1, i.e. 50% Lemon juice and 50% sugar syrup.

In sugar syrup, 100% is sugar syrup.

These two are mixed in the ratio 1:3.

$$\text{Lemon juice} = \frac{1(50\%)}{1+3}$$

$$\text{Sugar syrup} = \frac{1(50\%)+3(100\%)}{1+3} = \frac{350}{4}$$

Required ratio = 50:350 = 1:7

The answer is option D.

VIDEO SOLUTION

9.C

Let the alloy contain x Kg silver and y kg copper

Now when mixed with 3Kg Pure silver

$$\text{we get } \frac{(x+3)}{x+y+3} = \frac{9}{10}$$

we get $10x+30=9x+9y+27$

$$9y-x=3 \quad (1)$$

Now as per condition 2

silver in 2nd alloy = $2(0.9)=1.8$

$$\text{so we get } \frac{(x+1.8)}{x+y+2} = \frac{21}{25}$$

$$\text{we get } 21y-4x=3 \quad (2)$$

solving (1) and (2) we get $y=0.6$ and $x=2.4$

$$\text{so } x+y=3$$

 VIDEO SOLUTION

10. B

Considering the three kinds of tea are A, B, and C.

The price of kind A = Rs 800 per kg.

The price of kind B = Rs 500 per kg.

The price of kind C = Rs 300 per kg.

They were mixed in the ratio of 2 : 3: 5.

1/6 of the total mixture is sold for Rs 700 per kg.

Assuming the ratio of mixture to A = 12kg, B = 18kg, C =30 kg.

The total cost price is $800*12+500*18+300*30 = \text{Rs } 27600$.

Selling 1/6 which is 10kg for Rs 700/kg the revenue earned is Rs 7000.

In order to have an overall profit of 50 percent on Rs 27600.

Thes selling price of the 60 kg is $\text{Rs } 27600*1.5 = \text{Rs } 41400$.

Hence he must sell the remaining 50 kg mixture for $\text{Rs } 41400 - \text{Rs } 7000 = 34400$.

Hence the price per kg is $\text{Rs } 34400/50 = \text{Rs } 688$

 VIDEO SOLUTION



CAT Important Topics

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11. 45

Let initial volume be V, final be F for milk.

The formula is given by : $F = V \cdot \left(1 - \frac{K}{V}\right)^n$ n is the number of times the milk is drawn and replaced.

$$\text{so we get } F = V \left(1 - \frac{K}{V}\right)^2$$

here $K=9$

we get

$$\frac{16}{25}V = V \left(1 - \frac{9}{V}\right)^2$$

$$\text{we get } 1 - \frac{9}{V} = \frac{4}{5} \text{ or } -\frac{4}{5}$$

$$\text{If considering } 1 - \frac{9}{V} = -\frac{4}{5}$$

$V=5$, but this is not possible because 9 liters is drawn every time.

$$\text{Hence : } 1 - \frac{9}{V} = \frac{4}{5}, V = 45 \text{ liters}$$

 VIDEO SOLUTION

12. B

Let the selling price of the Large and Small boxes of chocolates be Rs.200 and Rs.100 respectively. Let us consider that the Large box has L grams of chocolate while the Small box has S grams of chocolate.

The relation between the selling price per gram of chocolate can be represented as: $\frac{200}{L} = 0.88 \times \frac{100}{S}$

On solving we obtain the ratio of the amount of chocolate in each box as: $\frac{L}{S} = \frac{25}{11}$

The percentage by which the weight of chocolate in the large box exceeds that in the small box = $\left(\frac{25}{11} - 1\right) \times 100 \approx 127\%$

 VIDEO SOLUTION

13. 200

Let Bottle A have an indigo solution of strength 33% while Bottle B have an indigo solution of strength 17%.

The ratio in which we mix these two solutions to obtain a resultant solution of strength 21% : $\frac{A}{B} = \frac{21-17}{33-21} = \frac{4}{12}$ or $\frac{1}{3}$

Hence, three parts of the solution from Bottle B is mixed with one part of the solution from Bottle A. For this process to happen, we need to displace 600 cc of solution from Bottle A and replace it with 600 cc of solution from Bottle B {since both bottles have 800 cc, three parts of this volume = 600cc}. As a result, 200 cc of the solution remains in Bottle B.

Hence, the correct answer is 200 cc.

 VIDEO SOLUTION

14. C

Initially let's consider A and B as one component

The volume of the mixture is doubled by adding A(60% alcohol) i.e they are mixed in 1:1 ratio and the resultant mixture has 72% alcohol.

Let the percentage of alcohol in component 1 be 'x'.

Using allegations, $\frac{(72-60)}{x-72} = \frac{1}{1} \Rightarrow x = 84$

Percentage of alcohol in A = 60% $\Rightarrow$ Let's percentage of alcohol in B = x%

The resultant mixture has 84% alcohol. ratio = 1:3

Using allegations, $\frac{(x-84)}{84-60} = \frac{1}{3}$

$\Rightarrow x = 92\%$

 VIDEO SOLUTION

15. B

Let the volume of Metals A,B,C be $3x, 4x, 7x$

Ratio weights of given volume be $5y, 2y, 6y$

$\therefore 15xy + 8xy + 42xy = 130 \Rightarrow 65xy = 130 \Rightarrow xy = 2$.

$\therefore$ The weight, in kg. of the metal C is $42xy = 84$.

 VIDEO SOLUTION



CAT WhatsApp Group



16.8

Initially the amount of Dye and Water are 16,24 respectively.

To make the ratio of Dye to Water to 2:5 the amount of water should be 40l for 16l of Dye=> 16l of water is added.

Now, the Dye and Water are 16,40 respectively.

After removing 1/4th of solution the amount of Dye and Water will be 12,30l respectively.

To have Dye and Water in the ratio of 2:3, for 30l of water we need 20l of Dye => 8l of Dye should be added.

Hence, 8 is correct answer.

 VIDEO SOLUTION

17.A

The weight/volume(g/L) for liquid 1 = 1000

The weight/volume(g/L) for liquid 2 = 800

The weight/volume(g/L) of the mixture = $480 / (1/2) = 960$

Using alligation the ratio of liquid 1 and liquid 2 in the mixture = $(960-800)/(1000-960) = 160/40 = 4:1$

Hence the percentage of liquid 1 in the mixture = $4 \times 100 / (4+1) = 80$

 VIDEO SOLUTION

18.D

Each of three vessels A, B, C contains 500 ml of salt solution of strengths 10%, 22%, and 32%, respectively.

The amount of salt in vessels A, B, C = 50 ml, 110 ml, 160 ml respectively.

The amount of water in vessels A, B, C = 450 ml, 390 ml, 340 ml respectively.

In 100 ml solution in vessel A, there will be 10ml of salt and 90 ml of water

Now, 100 ml of the solution in vessel A is transferred to vessel B. Then, 100 ml of the solution in vessel B is transferred to vessel C. Finally, 100 ml of the solution in vessel C is transferred to vessel A

i.e after the first transfer, the amount of salt in vessels A, B, C = 40, 120, 160 ml respectively.

after the second transfer, the amount of salt in vessels A, B, C = 40, 100, 180 ml respectively.

After the third transfer, the amount of salt in vessels A, B, C = 70, 100, 150 ml respectively.

Each transfer can be captured through the following table.

Salt solution		Initial Conc.(ml)	After 1st transfer	After 2nd transfer	After 3rd transfer
A	Total Conc.	500	400	400	500
	Salt Conc.	50	40	40	70
	Water	450	360	360	430
B	Total Conc.	500	600	500	500
	Salt Conc.	110	120	100	100
	Water	390	480	400	400
C	Total Conc.	500	500	600	500
	Salt Conc.	160	160	180	150
	Water	340	340	420	350

Percentage of salt in vessel A = $\frac{70}{500} \times 100$

=14%

VIDEO SOLUTION

19. C

Let the original number of students be 'n' whose average weight is 'x'

Let the number of students added be 'm' and the average weight will be $x + 3$

We need to find the value of $n : m$

It is given, average weight of students in a class increased by 0.6 after new students are added.

Therefore,

$$\frac{nx+m(x+3)}{n+m} = x + 0.6$$

$$nx + mx + 3m = mx + nx + 0.6n + 0.6m$$

$$2.4m = 0.6n$$

$$4m = n$$

$$\frac{n}{m} = \frac{4}{1}$$

The answer is option C.

VIDEO SOLUTION

20. A

Savings target in a year = 550×12 = Rs 6600

Saving in first 9 months = $9(4000-3500)$ = Rs 4500

Saving for remaining 3 months should be $6600-4500$, i.e. Rs 2100

Savings for each month in last 3 months = $\frac{2100}{3}$ = Rs 700

It is given, monthly expenses in last 3 months = Rs 3700

This implies, his monthly earnings from 10th month should be $3700+700$, i.e. Rs 4400

The answer is option A.

VIDEO SOLUTION



CAT Mock Test

Mocks designed exactly like the actual CAT



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21. 63

Let the number of students in section A and B be a and b, respectively.

It is given, $a = b - 10$

$\frac{32a+60b}{a+b}$ is an integer

$$\frac{32a+60(a+10)}{a+a+10} = k$$

$$\frac{46a+300}{a+5} = k$$

$$k = \frac{46(a+5)}{a+5} + \frac{70}{a+5}$$

$$k = 46 + \frac{70}{a+5}$$

a can take values 2, 5, 9, 30, 65

Difference = $65 - 2 = 63$

VIDEO SOLUTION

22. C

It is given that average of three numbers is 13.

$$\text{Sum} = 3 \times 13 = 39$$

It is given, $\frac{39+n}{4}$ is a odd number.

Minimum value $\frac{39+n}{4}$ can take such that n is a natural number is 11

$$\frac{39+n}{4} = 11$$

$$n = 5$$

The answer is option C.

VIDEO SOLUTION

23. D

$$a + 2b = 6$$

From the above equation, we can say that maximum value b can take is 3 and minimum value b can take is 0.

$$a + b + b = 6$$

$$a + b = 6 - b$$

a + b is maximum when b is minimum, i.e. b = 0

$$\text{Maximum value of } a + b = 6 - 0 = 6$$

a + b is minimum when b is maximum, i.e. b = 3

$$\text{Minimum value of } a + b = 6 - 3 = 3$$

$$\text{Average} = \frac{6+3}{2} = 4.5$$

The answer is option D.

VIDEO SOLUTION

24. C

$$\text{Given, } \frac{\text{sum of scores in } n \text{ matches} + 38 + 15}{n+2} = 29$$

$$\text{Given, } \frac{\text{sum of scores in } n \text{ matches}}{n} = 30$$

$$\Rightarrow 30n + 53 = 29(n+2) \Rightarrow n=5$$

$$\text{Sum of the scores in 5 matches} = 29 \times 7 - 38 - 15 = 150$$

Since the batsmen scored less than 38, in each of the first 5 innings. The value of x will be minimum when remaining four values are highest

$$\Rightarrow 37 + 37 + 37 + 37 + x = 150$$

$$\Rightarrow x = 2$$

VIDEO SOLUTION

25. 10

Let Total matches played be n and in initial n-10 matches his goals be x

$$\text{so we get } \frac{(x+1)}{n} = 0.15$$

$$\text{we get } x+1 = 0.15n \quad (1)$$

From condition (2) we get :

$$\frac{(x+2)}{n} = 0.2$$

we get $x+2 = 0.2n$ (2)
 Subtracting (1) and (2)
 we get $1 = 0.05n$
 $n = 20$
 So initially he played $n-10 = 10$ matches

 VIDEO SOLUTION



26. 92

Let sum of marks of students be x
 Now therefore $x = 25 \times 50 = 1250$
 Now to maximize the marks of the toppers
 We will minimize the marks of 20 students
 so their scores will be $(30, 31, 32, \dots, 49)$
 let score of toppers be y
 so we get $5y + \frac{20}{2}(79) = 1250$
 we get $5y + 790 = 1250$
 $5y = 460$
 $y = 92$
 So scores of toppers = 92

 VIDEO SOLUTION

27. B

Let us assume the family spends Rs. 100 each month for the first 3 months and then spends Rs. 50 in each of the next two months.

Then amount of onions bought = 10, 5, 4, 2, 1, for months 1-5 respectively.

Total amount bought = 22kg.

Total amount spent = $100 + 100 + 100 + 50 + 50 = 400$.

Average expense = $\frac{400}{22} = \text{Rs. } 18.18 \approx 18$

 VIDEO SOLUTION

28. D

The average marks for all the students is 38.

Sum = $5 \times 38 = 190$

To find the minimum marks scored by Amit, we need to maximise the score of remaining students.

Maximum scores sum of remaining students = $50 + 49 + 48 + 32 = 179$

Minimum possible score of Amit = $190 - 179 = 11$

It is given, Amit scored least. This implies maximum possible score of Amit is 31.

Difference = $31 - 11 = 20$

The answer is option D.

 VIDEO SOLUTION

29. C

It is given that the average of the 30 integers = 5

Sum of the 30 integers = $30 \times 5 = 150$

There are exactly 20 integers whose value is less than 5.

To maximise the average of the 20 integers, we have to assign minimum value to each of the remaining 10 integers

So the sum of 10 integers = $10 \times 6 = 60$

The sum of the 20 integers = $150 - 60 = 90$

Average of 20 integers = $\frac{90}{20} = 4.5$

 VIDEO SOLUTION

30. B

Total marks = $80 \times 10 = 800$

Total marks except highest and lowest marks = $81 \times 8 = 648$

So Summation of highest marks and lowest marks will be = $800 - 648 = 152$

When highest marks is 92, lowest marks will be = $152 - 92 = 60$

 VIDEO SOLUTION



31. A

Let the fixed income be x and the number of boarders be y .

$$x + 25y = 17500$$

$$x + 50y = 30000$$

$$\Rightarrow y = 500 \text{ and } x = 5000$$

$$x + 100y = 5000 + 50000 = 55000$$

$$\text{Average expense} = \frac{55000}{100} = \text{Rs. } 550.$$

 VIDEO SOLUTION

32. B

The first five numbers could be $n-2, n-1, n, n+1, n+2$. The next two number would then be, $n+3$ and $n+4$, in which case, the average of all the 7 numbers would be $\frac{(5n+2n+7)}{7} = n+1$

 VIDEO SOLUTION

33. B

Let x, y and z be no. of students in class X, Y, Z respectively.

From 1st condition we have

$$83x + 76y = 79x + 79y \text{ which give } 4x = 3y.$$

$$\text{Next we have } 76y + 85z = 81(y+z) \text{ which give } 4z = 5y.$$

Now overall average of all the classes can be given as $\frac{83x+76y+85z}{x+y+z}$

Substitute the relations in above equation we get,

$$\frac{83x+76y+85z}{x+y+z} = (83 \times \frac{3}{4} + 76 + 85 \times \frac{5}{4}) / (\frac{3}{4} + 1 + \frac{5}{4}) = 978/12 = 81.5$$

 VIDEO SOLUTION

34. B

Let the actual average be n . So, the new average is $n + 1.8$

Actual total = $10n$

New total = $10n + 18$

Let the number which was miswritten = ab (a is the tenth's digit and b is the units digit) = $10a+b$

and reversed number $ba = 10b+a$

So, $10b + a - (10a + b) = 18$

$\Rightarrow 9(b-a) = 18$

$\Rightarrow b-a = 2$

 VIDEO SOLUTION

35. E

Ten years ago, the total age of the family is 231 years.

Seven years ago, (Just before the death of the first person), the total age of the family would have been $231+8*3 = 231+24 = 255$.

This is because, in 3 years, every person in the family would have aged by 3 years,

Total change in age = $231+24 = 255$

After the death of one member, the total age is $255-60 = 195$ years.

Since a child takes birth in the same year, the number of members remain the same i.e. $(7+1) = 8$

Four years ago, (i.e. 6 years after start date) one of the member of age 60 dies, therefore, total age of the family is $195+24-60 = 159$ years.

Since a child takes birth in the same year, the number of members remain the same i.e. $(7+1) = 8$

After 4 more years, the current total age of the family is $= 8*4 + 159 = 191$ years

The average age is $191/8 = 23.875$ years = 24 years (approx)

Alternatively,

Since the number of members is always the same throughout

The 2 older members dropped their age by 60

So, after 10yrs, total age = $231 + 8*10 - 2*60 = 191$

Average age = $191/8 = 23.875 \simeq 24$

 VIDEO SOLUTION



CAT Syllabus PDF



36. B

The odd numbers in the set are 3, 5, 7, ..., $2n+1$

Sum of the odd numbers = $3+5+7+\dots+(2n+1) = n^2 + 2n$

Average of odd numbers = $n^2 + 2n/n = n+2$

Sum of even numbers = $2 + 4 + 6 + \dots + 2n = 2(1+2+3+\dots+n) = 2 \cdot n \cdot (n+1)/2 = n(n+1)$

Average of even numbers = $n(n+1)/n = n+1$

So, difference between the averages of even and odd numbers = 1

▶ VIDEO SOLUTION

37. A

Let, the average score of boys in the mid semester exam is A .

Therefore, the average score of girls in the mid semester exam be $A+5$.

Hence, the total marks scored by the class is $20 \times (A) + 30 \times (A+5) = 50 \times A + 150$

The average score of the entire class is $\frac{(50 \times A + 150)}{50} = A + 3$

wkt, class average increased by 2, class average in final term = $(A + 3) + 2 = A + 5$

Given, that score of girls dropped by 3, i.e $(A + 5) - 3 = A + 2$

Total score of girls in final term = $30 \times (A + 2) = 30A + 60$

	MID-TERM		FINAL-TERM	
No Boys/Girls	Avg score	Total score	Avg score	Total score
20	A	$20A$		
30	$A+5$	$30A+150$	$A+2$	$30A+60$
Class score		$50A+150$		
Class Average		$A+3$		$A+5$

Total class score in final term = $(A + 5) \times 50 = 50A + 250$

the total marks scored by the boys is $(50A + 250) - (30A - 60) = 20A + 190$

Hence, the average of the boys in the final exam is $\frac{(20G + 190)}{20} = A + 9.5$

	MID-TERM		FINAL-TERM	
No Boys/Girls	Avg score	Total score	Avg score	Total score
20	A	$20A$	$A+9.5$	$20A+190$
30	$A+5$	$30A+150$	$A+2$	$30A+60$
Class score		$50A+150$		$50A+250$
Class Average		$A+3$		$A+5$

Hence, the increase in the average marks of the boys is $(A + 9.5) - A = 9.5$

▶ VIDEO SOLUTION

38. C

Let the average height of 22 toddlers be $3x$.

Sum of the height of 22 toddlers = $66x$

Hence average height of the two toddlers who left the group = x

Sum of the height of the remaining 20 toddlers = $66x - 2x = 64x$

Average height of the remaining 20 toddlers = $64x/20 = 3.2x$

Difference = $0.2x = 2$ inches $\Rightarrow x = 10$ inches

Hence average height of the remaining 20 toddlers = $3.2x = 32$ inches

 VIDEO SOLUTION

39.11

It is given that the maximum weight limit is 630. The lightest person's weight is 53 Kg and the heaviest person's weight is 57 Kg.

In order to have maximum people in the lift, all the remaining people should be of the lightest weight possible, which is 53 Kg.

Let there be n people.

$$53 + n(53) + 57 = 630$$

n is approximately equal to 9.8. Hence, 9 people are possible.

Therefore, a total of $9 + 2 = 11$ people can use the elevator.

 VIDEO SOLUTION

40.60

Let the total number of tests be ' n ' and the average by ' A '

Total score = $n \cdot A$

When 1st 10 tests are excluded, decrease in total value of scores = $(nA - 20 \cdot 10) = (nA - 200)$

Also, $(n - 10)(A + 1) = (nA - 200)$

On solving, we get $10A - n = 190$(i)

When last 10 tests are excluded, decrease in total value of scores = $(nA - 30 \cdot 10) = (nA - 300)$

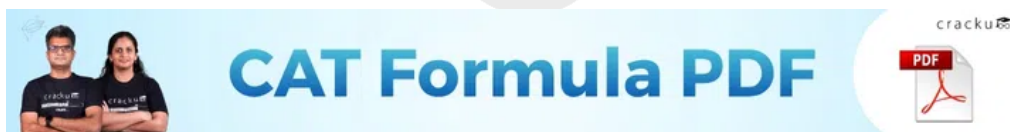
Also, $(n - 10)(A - 1) = (nA - 300)$

On solving, we get $10A + n = 310$(ii)

From (i) and (ii), we get $n = 60$

Hence, 60 is the correct answer.

 VIDEO SOLUTION



41.D

The possible average age of people whose ages are below 51 years will be maximum if the average age of the number of people aged 51 years and above is minimum. Hence, we can say that there are 30 people having same age 51 years.

Let ' x ' be the maximum average age of people whose ages are below 51.

Then we can say that,

$$\frac{51 \cdot 30 + 39 \cdot x}{30 + 39} = 38$$

$$\Rightarrow 1530 + 39x = 2622$$

$$\Rightarrow x = 1092/39 = 28$$

Hence, we can say that option D is the correct answer.

 VIDEO SOLUTION

42. B

Assume the average of 21 students other than Ramesh = a

Sum of the scores of 21 students other than Ramesh = $21a$

Hence the average of 22 students = $a+1$

Sum of the scores of all 22 students = $22(a+1)$

The score of Ramesh = Sum of scores of all 22 students - Sum of the scores of 21 students other than Ramesh
 $= 22(a+1) - 21a = a+22 = 82.5$ (Given)

$\Rightarrow a = 60.5$

Hence, sum of the scores of all 22 students = $22(a+1) = 22 \times 61.5 = 1353$

Now the sum of the scores of students other than Gautam = $21 \times 62 = 1302$

Hence the score of Gautam = $1353 - 1302 = 51$

 VIDEO SOLUTION

43. 15

Let us assume the four numbers to be a, b, c and d in ascending order.

Average of first two numbers is 1 more than the first number

$$\frac{(a+b)}{2} = a + 1$$

$$b - a = 2$$

$$b = a + 2$$

Average of first three numbers is 2 more than average of first two numbers

$$\frac{(a+b+c)}{3} = \frac{(a+b)}{2} + 2$$

$$2c = a + b + 12$$

Substituting the value for b

$$2c = a + a + 2 + 12$$

$$2c = 2a + 14$$

$$c = a + 7$$

Average of first four numbers is 3 more than average of first three numbers.

$$\frac{(a+b+c+d)}{4} = \frac{(a+b+c)}{3} + 3$$

$$3d = a + b + c + 36$$

Substituting the value of b and c

$$3d = a + a + 2 + a + 7 + 36$$

$$3d = 3a + 45$$

$$d = a + 15$$

d is the largest and a is the smallest and we know that $d = a + 15$

Hence the difference between the smallest and the largest values is 15.

 VIDEO SOLUTION

44. 70

We are given that average of three distinct integers is 28, that means the sum of these three integers is $28 \times 3 = 84$

Let us write $x + y + z = 84$

x, y, z being the three distinct integers in ascending order.

If the smallest number is increased by 7 and the largest number is reduced by 10

$$(x + 7) + (y) + (z - 10) = 81$$

New arithmetic mean will be $\frac{81}{3} = 27$

And this is said to be 2 more than the middle number, meaning

$$27 - 2 = y = 25$$

$$x + z = 59$$

We are given that difference between the largest and the smallest numbers becomes 64,

$$(z - 10) - (x + 7) = 64$$

$$z - x = 81$$

Adding the two equations we get, $2z = 140$

$$z = 70$$

 VIDEO SOLUTION

45. C

Let the average salary of managers be x and let the average salary of engineers be y . The total salary of all the employees will be $(5 + 25) * 60000 = 1800000$

$$\text{We have, } 5x + 25y = 1800000 \dots (1)$$

If the average salary of all the employees increases by 5%, the total salary of all the employees will also increase by 5%, because the total number of employees remains the same. The new total salary will be, $1800000 \times 1.05 = 1890000$

Also, the average salary of all the managers has increased by 20% and has become $1.2x$, we have

$$5 * 1.2x + 25y = 1890000, \text{ or } 6x + 25y = 1890000 \dots (2)$$

Subtracting equation (1) from equation (2), we get,

$$x = 90000$$

$$\text{Which gives } 25y = 1800000 - 90000 * 5 \text{ or } y = \frac{1350000}{25} = 54000$$

Therefore, the correct answer is option C.

 VIDEO SOLUTION



46. 49

Let the copies sold on days 1 through 9 be $d_1, \dots, d_9$

From the averages:

$$d_1 + \dots + d_7 = 7 \times 60 = 420$$

$$d_1 + \dots + d_8 = 8 \times 63 = 504$$

$$\text{So } d_8 = 504 - 420 = 84. \text{ Then } d_9 = 84 - 11 = 73$$

$$\text{The sum from day 2 to day 9 is } 8 \times 66 = 528$$

$$d_2 + \dots + d_9 = (d_1 + \dots + d_8) - d_1 + d_9 = 504 - d_1 + 73 = 577 - d_1$$

$$\text{Thus } 577 - d_1 = 528 \Rightarrow d_1 = 49$$

 VIDEO SOLUTION

47. B

Let us say the capacity of the glass is X, and it is completely filled with milk,

If two-thirds of its content is poured out and replaced with water, the remaining fraction of the milk will be one third.

And this is said to be done three more times, that means a total of 4 times.

So the contents of Milk initially being X,

And after 4 times the contents will be, $X \left(1 - \frac{2}{3}\right)^4 = \frac{X}{81}$

Since the total contents is X, and the milk contents is X/81, the water contents will be 80X/81.

Ratio of milk to water will be $\frac{X}{81} : \frac{80X}{81}$

Answer is 1 : 80

 VIDEO SOLUTION

48. B

Let's start from the step when there was 50% concentration.

Let's take there to be 2T solution: T acid and T water.

Adding 15 litres of acid increases the acid concentration to 80%, giving the equation $\frac{T+15}{2T+15} = \frac{4}{5}$

Solving this would give us T=5

This means that there were 5 litres of acid and 5 litres of water after mixing 2 litres of water.

Therefore, there would be 5 litres of acid and 3 litres of water before adding the water.

We are asked the ratio of water to acid, which would be 3:5

Therefore, Option B is the correct answer.

 VIDEO SOLUTION

49. B

We are told that Rajesh manages 20 hectares and Vimal manages 30 hectares

For Vimal, we know the distribution of the land between Wheat and Mustard, 5:3

So, wheat area will be, $\frac{5}{8} (30)$

Mustard area will be, $\frac{3}{8} (30)$

Similarly, let us assume that the distribution of crops between Wheat and Mustard to be k:1

Wheat will be, $\frac{k}{k+1} (20)$

Mustard will be, $\frac{1}{k+1} (20)$

We are told that total area of Wheat and Mustard is in the ratio 11:9

Adding them up we get,

$$\frac{\left(\frac{150}{8} + \frac{20k}{k+1}\right)}{\left(\frac{90}{8} + \frac{20}{k+1}\right)} = \frac{11}{9}$$

$$\frac{\left(\frac{15}{8} + \frac{2k}{k+1}\right)}{\left(\frac{9}{8} + \frac{2}{k+1}\right)} = \frac{11}{9}$$

$$\frac{135}{8} + \frac{18k}{k+1} = \frac{99}{8} + \frac{22}{k+1}$$

$$\frac{36}{8} = \frac{22-18k}{k+1}$$

$$44 - 36k = 9k + 9$$

$$45k = 35$$

$$k = \frac{7}{9}$$

Hence the ratio of distribution of area between Wheat and Mustard for Rajesh is $\frac{7}{9}$

VIDEO SOLUTION

50.30

Let us assume the amount of Milk in the container to be X and the amount of water in the container to be Y . We are told that $X+Y=300$.

Now, an operation is given where "an amount of this solution was taken out from the container which was twice the volume of water that was earlier poured into it, and water was poured to refill the container again"

Volume of the water initially is Y . If twice that amount is taken out, the percentage of the solution that is taken out will be, $\frac{2Y}{X+Y}$

That means the quantity of milk that will remain in the solution will be, $X \left(1 - \frac{2Y}{X+Y}\right)$

This value is given to be 72%, 72% of 300 will be 216

$$X \left(1 - \frac{2Y}{X+Y}\right) = 216$$

$$X \left(\frac{X+Y-2Y}{X+Y}\right) = 216$$

Writing $X = 300 - Y$

$$(300 - Y)(300 - 2Y) = 64800$$

Expanding this we have,

$$2Y^2 - 900Y + 25200 = 0$$

Factorising this equation we have,

$$2(Y - 30)(Y - 420) = 0$$

Y is either 30 or 420.

Given that the capacity of the container itself is 300, Y has to be 30.

Hence the amount of water initially is 30 Litres.

VIDEO SOLUTION

CAT Percentile Predictor



51.16

Vessel A contains 60 litres of pure alcohol, and Vessel B contains 60 litres of pure water. Let the amount taken out from vessel A be x litres. The contents in Vessel A now are $60 - x$ litres of alcohol, and in Vessel B now are 60 litres of water and x litres of alcohol.

After mixing, when the same quantity x is taken out from Vessel B, the alcohol that would be taken out would

be $\left(\frac{x}{60+x}\right)^{\text{th}}$ of x , which is $\frac{x^2}{60+x}$ litres.

The initial total quantity of 60 litres has been restored in A after replacement. The total amount of alcohol in Vessel A after the process is complete is:

$$60 - x + \frac{x^2}{60+x} = \frac{(60+x)(60-x) + x^2}{60+x} = \frac{3600}{60+x}$$

The total quantity in vessel A is 60 litres where the ratio of alcohol to total quantity is $\frac{15}{15+4} = \frac{15}{19}$

Therefore,

$$\frac{3600 \div (60+x)}{60} = \frac{15}{19}$$

$$\Rightarrow \frac{60}{60+x} = \frac{15}{19}$$

$$\Rightarrow x = 16$$

Thus, the quantity taken out and replaced in the two iterations is 16 litres.

 VIDEO SOLUTION

52. A

Let coffee price = C Rs/kg and cocoa price = K Rs/kg.

From the two given mixtures:

$$0.16C + 0.84K = 240$$

$$0.36C + 0.64K = 320$$

Multiply both equations by 100 to remove decimals:

$$16C + 84K = 24000$$

$$36C + 64K = 32000$$

Subtract the first from the second:

$$(36C + 64K) - (16C + 84K) = 32000 - 24000$$

$$20C - 20K = 8000 \Rightarrow C - K = 400$$

Put $C = K + 400$ into $16C + 84K = 24000$:

$$16(K + 400) + 84K = 24000$$

$$16K + 6400 + 84K = 24000$$

$$100K = 17600 \Rightarrow K = 176$$

So $C = 176 + 400 = 576$ (Rs/kg).

For the new mixture priced at Rs 376/kg, let the coffee fraction be p . Then

$$p \cdot 576 + (1 - p) \cdot 176 = 376$$

Upon solving $p = \frac{1}{2}$

Thus, coffee is (50%) of the new mixture. In 10 kg of this mixture, coffee = $10 \times 0.5 = 5\text{kg}$

 VIDEO SOLUTION

53. D

200 L with (30%) acid $\Rightarrow$ acid = $0.30 \times 200 = 60\text{L}$

Replace 20% with water: remaining acid = $60 \times (1 - 0.20) = 60 \times 0.8 = 48\text{L}$.

Replace 10% with pure acid: After removing 10% of the mixture, the acid becomes $48 \times 0.9 = 43.2\text{L}$, then adding back 20 L pure acid $\Rightarrow$ acid = $43.2 + 20 = 63.2\text{L}$.

Replace 15 with water: acid left = $63.2 \times (1 - 0.15) = 63.2 \times 0.85 = 53.72\text{L}$.

Final concentration = $\frac{53.72}{200} = 0.2686 \approx 26.86\%$.

The answer is 27%

 VIDEO SOLUTION